

6.4 Maximum Likelihood and Least-Squared Error Hypotheses – Notes

1. Motivation

- Many learning algorithms (linear regression, polynomial curve fitting, neural networks) minimize **sum of squared errors (SSE)** between predictions and training labels.
 - Question: Why squared error? Is it arbitrary?
 - **Bayesian justification:** Under Gaussian noise, minimizing SSE = choosing the **Maximum Likelihood (ML) hypothesis**.
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2. Problem Setting

- Hypothesis space H : real-valued functions

$$h : X \rightarrow \mathbb{R}$$

- Target function: $f : X \rightarrow \mathbb{R}$.
- Training data:

$$D = \{(x_1, d_1), \dots, (x_m, d_m)\}, \quad d_i = f(x_i) + \epsilon_i$$

- Noise assumption:
 - $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$
 - Independent across examples.
 - Goal: find hypothesis h_{ML} that maximizes likelihood of data.
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3. Likelihood of Data

- For a single point:

$$p(d_i | h(x_i)) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(d_i - h(x_i))^2}{2\sigma^2}\right)$$

- For all m examples (independent):

$$p(D | h) = \prod_{i=1}^m p(d_i | h(x_i))$$

4. Log-Likelihood

- Take log to simplify:

$$\ell(h) = \log p(D | h) = -\frac{m}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^m (d_i - h(x_i))^2$$

- First term is constant wrt h .
- So maximizing log-likelihood = **minimizing squared error**:

$$h_{ML} = \arg \min_{h \in H} \sum_{i=1}^m (d_i - h(x_i))^2$$

5. Key Result

- **Least-Squares Hypothesis = Maximum Likelihood Hypothesis** (under Gaussian noise).
 - Squared error minimization is not arbitrary but probabilistically justified.
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6. Why Normal Distribution?

- Convenient mathematically.
 - Justified by **Central Limit Theorem**: sum of many independent small noise sources $\approx$ Gaussian.
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7. Generalization to Other Noise Models

- Noise distribution determines the loss:
 - **Gaussian noise** $\rightarrow$ Squared Error (L2).
 - **Laplace noise** $\rightarrow$ Absolute Error (L1).
 - **Uniform noise** $\rightarrow$ Minimize maximum error.
 - Thus, the choice of loss encodes an assumption about the noise.
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8. Practical Implications

- **Linear Regression**: OLS = ML under Gaussian noise.
 - **Neural Networks**: SSE loss = ML under Gaussian output noise.
 - **Curve fitting**: Polynomial least squares = ML.
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9. Limitations

- Assumes noise only in **outputs** d_i , not in **inputs** x_i .
 - Real-world noise can be non-Gaussian, correlated, or affect inputs.
 - In such cases, squared error $\neq$ ML $\rightarrow$ need robust losses or explicit noise modeling.
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10. Teaching Tips

- Draw **Figure 6.2**: true function (solid line), noisy samples (points), least-squares fit (dashed line).
 - Derive log-likelihood step by step, show exponential becomes squared error.
 - Extend discussion: Laplace noise $\rightarrow$ L1, show generality.
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 **Takeaway:** Minimizing squared error is equivalent to Maximum Likelihood estimation under Gaussian noise — a powerful justification for its widespread use.